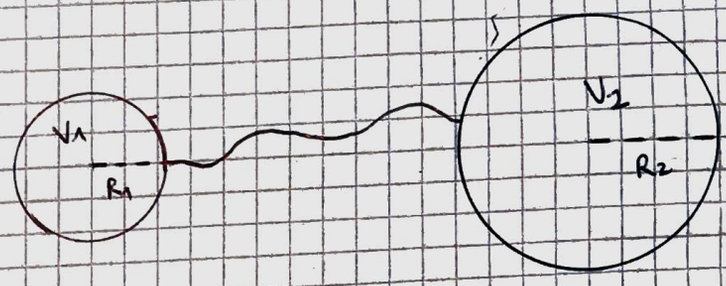
APPELLO 7/07/2019
ESERCIZIO 1



la carica iniziale sui due corpi è:

$Q_1 = 4\pi\varepsilon_0 R_1 V_1$

$Q_2 = 4\pi\varepsilon_0 R_2 V_2$

a) b)

$V_{1f} = V_{2f}$

$Q_{1f} + Q_{2f} = Q_1 + Q_2$

$\Rightarrow \cancel{4\pi\varepsilon_0} R_1 V_{1f} + \cancel{4\pi\varepsilon_0} R_2 V_{2f} = \cancel{4\pi\varepsilon_0} R_1 V_1 + \cancel{4\pi\varepsilon_0} R_2 V_2$

$$V_{1f} = V_{2f} = \frac{R_1 V_1 + R_2 V_2}{R_1 + R_2}$$

$$Q_{1f} = 4\pi\varepsilon_0 R_1 \frac{R_1 V_1 + R_2 V_2}{R_1 + R_2}$$

$$Q_{2f} = 4\pi\varepsilon_0 R_2 \frac{R_1 V_1 + R_2 V_2}{R_1 + R_2}$$

$\Delta Q = Q_1 - Q_{1f} = 4\pi\varepsilon_0 R_1 R_2 (V_1 + V_2)$

è questo

$\cancel{P_J = (V_1 - V_2)\Delta Q = 4\pi\varepsilon_0 R_1 R_2 (V_1^2 - V_2^2)}$

$$\cancel{P_J = (V_{1f} - V_1)\Delta Q = \frac{4\pi\varepsilon_0 R_1 R_2^2 (V_2 - V_1)}{R_1 + R_2}}$$

$P_J = (V_1 - V_2)\Delta Q = 4\pi\varepsilon_0 R_1 R_2 (V_1^2 - V_2^2)$

ESERCIZIO 4

$\lambda_1 = 500 \text{ nm}$

$\lambda_2 = 501 \text{ nm}$

$L = 5 \text{ cm}$

$$R = \frac{\lambda_1}{\lambda_2 - \lambda_1} = nN \Rightarrow N = \frac{\lambda_1}{\lambda_2 - \lambda_1} = 500$$

b) $D = 1 \text{ m}$

$$d \operatorname{sen} \vartheta^{MAX} = n\lambda_1$$

$$\frac{L}{N} \frac{x_n^{MAX}}{D} = n\lambda_1$$

$$x_1^{MAX} = \frac{\lambda_1 N D}{L}$$